



Programa de Pós-graduação em
INFORMÁTICA



PUC Minas



Teoria dos Grafos e Computabilidade

— Sets on graphs —
(Professor version)

Silvio Jamil F. Guimarães

Graduate Program in Informatics – PPGINF

Image and Multimedia Data Science Laboratory – IMScience

Pontifical Catholic University of Minas Gerais – PUC Minas



Programa de Pós-graduação em
INFORMÁTICA



PUC Minas



Teoria dos Grafos e Computabilidade

— Independent sets —
(Professor version)

Silvio Jamil F. Guimarães

Graduate Program in Informatics – PPGINF

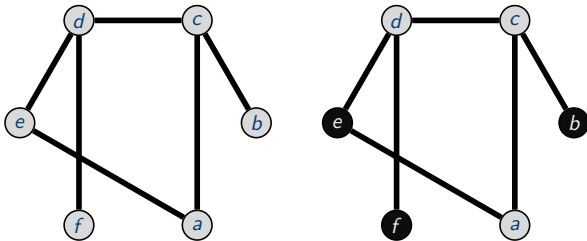
Image and Multimedia Data Science Laboratory – IMScience

Pontifical Catholic University of Minas Gerais – PUC Minas

Independent sets

Let $G = (V, E)$ be an undirected connected graph.

- ▶ A subset $S \subseteq V$ is an **independent set** if $\forall u, v \in S$ there **is not** ~~exist~~ an edge $(u, v) \in E$.
- ▶ Independent sets have also been called internally stable sets.



Independent sets

Let $G = (V, E)$ be an undirected connected graph, and S an independent set of G

- ▶ We say that the subset $S \subseteq V$ is a **maximal independent set** if there is no other independent set A in which $S \subset A$;
- ▶ The number of **internal stability** $\beta(G)$ is equal to the cardinality of the largest maximal independent set.

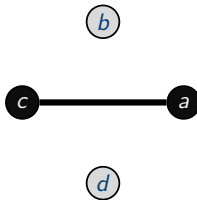
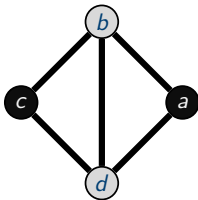
As S is an independent set of G , then S is a **clique** in the complement graph.

Independent sets

Let $G = (V, E)$ be an undirected connected graph, and S an independent set of G

- ▶ We say that the subset $S \subseteq V$ is a **maximal independent set** if there is no other independent set A in which $S \subset A$;
- ▶ The number of **internal stability** $\beta(G)$ is equal to the cardinality of the largest maximal independent set.

As S is an independent set of G , then S is a **clique** in the complement graph.



Independent sets

Let $G = (V, E)$ be an undirected connected graph. **Design** a method for computing an independent set of G

Algorithm: A method for computing an independent set

input : A graph $G = (V, E)$.

output: A independent set S

```
1  $S = \emptyset$ ;  
2 while  $V \neq \emptyset$  do  
3    $u =$  vertex with the smallest degree in  $G$ ;  
4    $V = V - \{u\} - \Gamma(u)$ ;  
5    $S = S \cup \{u\}$ ;  
6 end  
7 return  $S$ ;
```

Questions?

Sets on graphs
– Independent sets –



Programa de Pós-graduação em
INFORMÁTICA



PUC Minas



Teoria dos Grafos e Computabilidade

— Dominating sets —
(Professor version)

Silvio Jamil F. Guimarães

Graduate Program in Informatics – PPGINF

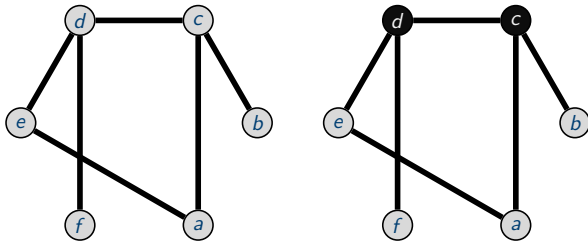
Image and Multimedia Data Science Laboratory – IMScience

Pontifical Catholic University of Minas Gerais – PUC Minas

Dominating sets

Let $G = (V, E)$ be an undirected connected graph.

- ▶ A subset $S \subseteq V$ is an **dominating set** if $\forall u \in S$ there exist a $v \in V - S$ such that $(u, v) \in E$.
- ▶ Dominating sets have also been called externally stable sets.



Let $G = (V, E)$ be an undirected connected graph, and S a dominating set of G

- ▶ We say that the subset $S \subseteq V$ is a **minimal dominating set** if there is no other dominating set A in which $A \subset S$;
- ▶ The number of **external stability** $\beta(G)$ is equal to the cardinality of the smallest minimal dominating set.

Independent sets

Let $G = (V, E)$ be an undirected connected graph. **Design** a method for computing a dominance set of G

Algorithm: A method for computing a dominating set

input : A graph $G = (V, E)$.

output: A dominating set D

```
1  $D = \emptyset$ ;  
2 while  $V \neq \emptyset$  do  
3    $u =$  vertex with the highest degree in  $G$ ;  
4    $V = V - \{u\} - \Gamma(u)$ ;  
5    $D = D \cup \{u\}$ ;  
6 end  
7 return  $D$ ;
```

Questions?

Sets on graphs
– Dominating sets –



Programa de Pós-graduação em
INFORMÁTICA



PUC Minas



Teoria dos Grafos e Computabilidade

— Vertex cover —
(Professor version)

Silvio Jamil F. Guimarães

Graduate Program in Informatics – PPGINF

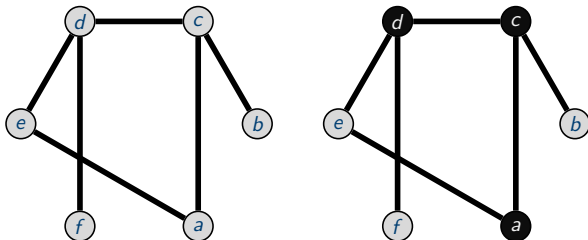
Image and Multimedia Data Science Laboratory – IMScience

Pontifical Catholic University of Minas Gerais – PUC Minas

Vertex cover

Let $G = (V, E)$ be an undirected connected graph.

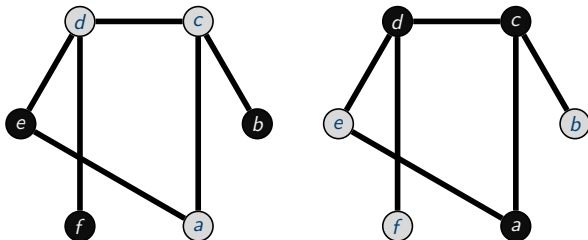
- A subset $S \subseteq V$ is an **vertex cover** if $\forall (u, v) \in E$, either $u \in S$ or $v \in S$.



Vertex cover

Let $G = (V, E)$ be an undirected connected graph, and S a vertex cover of G

As S is a vertex cover of G , then $V-S$ is an independent set.



Let $G = (V, E)$ be an undirected connected graph. **Design** a method for computing a vertex cover in G

Algorithm: A method for computing a minimum vertex cover

input : A graph $G = (V, E)$.

output: A ~~independent set~~ S
vertex cover

```
1  $S = \emptyset$ ;  
2 while  $E \neq \emptyset$  do  
3   Let  $(u, v)$  an arbitrary edge of  $E$ ;  
4   Choose either  $u$  or  $v$  to be included to  $S$ ;  
5    $S = S \cup \{u\}$  for instance;  
6    $V = V - u$ ;  
7 end  
8 return  $S$ ;
```

Questions?

Sets on graphs
– Vertex cover –